

A PRIME-POWER CONGRUENCE FOR AN ODD-EXPONENT BINOMIAL FAMILY

ABSTRACT. For a nonnegative integer m , define

$$u_m(n) = \text{Num} \left(\sum_{k=1}^n \frac{1}{k^{2m+1}} \binom{n}{k}^2 \binom{n+k}{k}^2 \right),$$

where the numerator is taken after reducing the rational number to lowest terms. The conjecture treated here originates in the record for [OEIS A357513](#), which states this parameterized form.

We determine exactly when p^4 divides $u_m(p-1)$. For every prime p ,

$$p^4 \mid u_m(p-1) \iff p-1 \nmid 2m+4 \text{ or } p \mid 2m+7.$$

1. STATEMENT

For a fixed nonnegative integer m , put

$$\mathcal{E}_m = \{p \text{ prime} : p-1 \mid 2m+4 \text{ and } p \nmid 2m+7\}.$$

Theorem 1. *Let $m \geq 0$ be an integer and let p be prime. Then*

$$p^4 \mid u_m(p-1) \iff p \notin \mathcal{E}_m.$$

If $p \in \mathcal{E}_m$, then

$$v_p(u_m(p-1)) = \begin{cases} 2, & p = 2, \\ 3, & p \text{ is odd.} \end{cases}$$

2. COMMON DENOMINATORS

Fix m , and write

$$d = 2m + 1.$$

For a prime p , let

$$S_{m,p} = \sum_{k=1}^{p-1} \frac{1}{k^d} \binom{p-1}{k}^2 \binom{p-1+k}{k}^2.$$

When p is odd, we work in the local ring

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, p \nmid b \right\}.$$

The reduced denominator of $S_{m,p}$ is coprime to p . Indeed, set

$$D_m = ((p-1)!)^d$$

and

$$N_m = \sum_{k=1}^{p-1} \left(\binom{p-1}{k} \binom{p-1+k}{k} \right)^2 \frac{D_m}{k^d}.$$

Since $k \mid (p-1)!$ for $1 \leq k \leq p-1$, every D_m/k^d is an integer. Thus $N_m \in \mathbb{Z}$, $S_{m,p} = N_m/D_m$, and $p \nmid D_m$.

Suppose that $S_{m,p} = A/B$ in lowest terms. Cross-multiplication gives $AD_m = N_mB$. Since $\gcd(A, B) = 1$, Euclid's lemma gives $B \mid D_m$. In particular, $p \nmid B$. Thus both B and D_m are p -adic units, so for every $r \geq 0$,

$$p^r \mid A \iff S_{m,p} \equiv 0 \pmod{p^r}. \quad (1)$$

In particular, a congruence for $S_{m,p}$ transfers directly to the reduced numerator $u_m(p-1)$.

3. THE MODULAR REDUCTION

For $1 \leq k \leq p-1$, the exact identities

$$\begin{aligned} \binom{p-1}{k} &= (-1)^k \prod_{j=1}^k \left(1 - \frac{p}{j}\right), \\ \binom{p-1+k}{k} &= \frac{p}{k} \prod_{j=1}^{k-1} \left(1 + \frac{p}{j}\right) \end{aligned}$$

give

$$\binom{p-1}{k} \binom{p-1+k}{k} = (-1)^k \frac{p}{k} \left(1 - \frac{p}{k}\right) \prod_{j=1}^{k-1} \left(1 - \frac{p^2}{j^2}\right). \quad (2)$$

After squaring (2) and dividing by k^d , the k th summand of $S_{m,p}$ is

$$\frac{p^2}{k^{d+2}} \left(1 - \frac{p}{k}\right)^2 \prod_{j=1}^{k-1} \left(1 - \frac{p^2}{j^2}\right)^2.$$

All denominators are units in $\mathbb{Z}_{(p)}$, and

$$\prod_{j=1}^{k-1} \left(1 - \frac{p^2}{j^2}\right)^2 \equiv 1 \pmod{p^2}.$$

The displayed summand already contains p^2 . Hence, modulo p^4 , it is congruent to

$$\frac{p^2}{k^{d+2}} - \frac{2p^3}{k^{d+3}}.$$

Put

$$r = d + 2 = 2m + 3, \quad e = d + 3 = 2m + 4$$

and define

$$H_s = \sum_{k=1}^{p-1} \frac{1}{k^s} \in \mathbb{Z}_{(p)}.$$

Summing the preceding congruence gives

$$S_{m,p} \equiv p^2 H_r - 2p^3 H_e \pmod{p^4}. \quad (3)$$

The exponent r is odd. Since $k \mapsto p-k$ permutes $1, \dots, p-1$,

$$\begin{aligned} \frac{1}{(p-k)^r} &= -\frac{1}{k^r} \left(1 - \frac{p}{k}\right)^{-r} \\ &\equiv -\frac{1}{k^r} - \frac{rp}{k^{r+1}} \pmod{p^2}. \end{aligned}$$

Because $r+1 = e$, summing over k yields

$$2H_r + rpH_e \equiv 0 \pmod{p^2}. \quad (4)$$

Multiply (4) by p^2 and double (3). Since $r + 4 = 2m + 7$, we obtain

$$2S_{m,p} \equiv -(2m + 7)p^3 H_e \pmod{p^4}. \quad (5)$$

4. THE FINITE-FIELD OBSTRUCTION

Reduction modulo p and the fact that inversion permutes $\mathbb{F}_p^\times$ give

$$H_e \equiv \sum_{x \in \mathbb{F}_p^\times} x^{-e} = \sum_{x \in \mathbb{F}_p^\times} x^e \pmod{p}.$$

The standard power-sum evaluation in the cyclic group $\mathbb{F}_p^\times$ is

$$H_e \equiv \begin{cases} 0 & \pmod{p}, & p - 1 \nmid e, \\ -1 & \pmod{p}, & p - 1 \mid e. \end{cases} \quad (6)$$

Assume first that p is odd. Then 2 is a unit in $\mathbb{Z}_{(p)}$. If $p - 1 \nmid e$, equations (5) and (6) give $S_{m,p} \equiv 0 \pmod{p^4}$. If $p - 1 \mid e$, those equations instead give

$$2S_{m,p} \equiv (2m + 7)p^3 \pmod{p^4}.$$

It follows that

$$S_{m,p} \equiv 0 \pmod{p^4} \iff p \mid 2m + 7.$$

Together with (1), this proves the asserted classification for every odd prime.

For $p = 2$, the sum has one term:

$$S_{m,2} = \binom{1}{1}^2 \binom{2}{1}^2 = 4.$$

Thus $u_m(1) = 4$, so $2^4 \nmid u_m(1)$ and $v_2(u_m(1)) = 2$. This agrees with the criterion because $2 - 1 \mid 2m + 4$, whereas $2 \nmid 2m + 7$.

It remains to record the valuation at an odd exceptional prime. Equation (4) shows that $p \mid H_r$ in $\mathbb{Z}_{(p)}$. Equation (3) therefore gives $p^3 \mid S_{m,p}$. On the other hand, if $p - 1 \mid 2m + 4$ and $p \nmid 2m + 7$, then (5) and (6) show that $S_{m,p} \not\equiv 0 \pmod{p^4}$. Consequently

$$v_p(u_m(p - 1)) = 3.$$

This completes the proof of Theorem 1.